University of Glasgow

ENG3034: Oscilloscope and Temperature Measurement Sensors

Lab Report

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Abstract

This report presents the results of the experiment of temperature measurements by using RTD, thermopile, and floating thermopile sensor devices by using an oscilloscope. The results are analyzed by using the SNR method for various levels of averaging, noise spectra by using the FFT method, bandwidth limiting, capacitors for passive filtering, and quantization by using the sensitivity method.

In the sensor performance analysis section, the results of the heating and cooling of the RTD sensor are compared with the thermopile sensor by using the error, sensitivity, RTD linearity error, RTD random uncertainty at room temperature. Then the results of the floating thermocouple sensor are compared with the RTD sensor by using the sensitivity, measurement error. Then the results of the sensor are analyzed by using the open circuit [disconnected sensor] method.

1 Experimental Setup

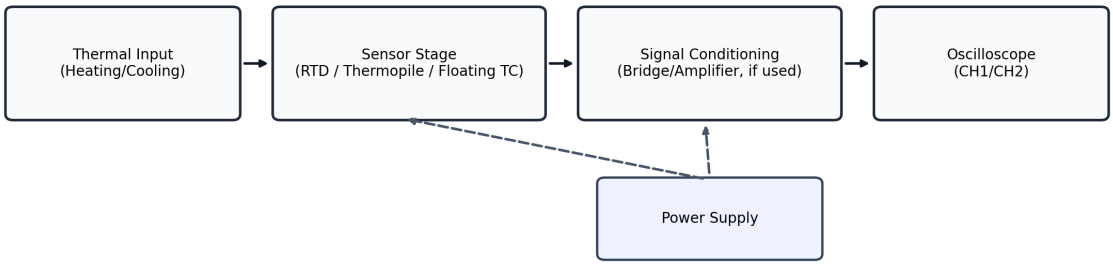


Figure 1: Main block diagram of the temperature measurement experiment.

Figure [1] illustrates the main measurement chain experiments parts: a heat or cooling input is provided to the sensor setup the signal is conditioned if needed and it is then measured by the oscilloscope channels. This same hardware chain was used throughout all the lab tests, varying only the sensor type and the measurement parameters such as averaging ,bandwidth limit and capacitor filtering.

2 Exercise 1: Noise, FFT, Filtering, Sampling, and Quantization

This section examines the measurement chain's quality prior to interpreting the results of a comparison between sensors. It entails SNR vs. averaging, FFT based noise analysis,

bandwidth limit observation, capacitor filtering effect, sampling rate selection and quantization error under sensitivity settings. It seeks to determine the impact of acquisition settings on noise, resolution and accuracy of the received temperature signal.

2.1 SNR Equation

We used:

$$\mathrm{SNR}_{\mathrm{csv,linear}} = \frac{\mu}{\sigma}[1] \tag{1}$$

$$\mathrm{SNR}_{dB} = 20\log_{10}(\mathrm{SNR}_{\mathrm{linear}})\,[1] \tag{2}$$

Computed SNR Averaging values

$$\mathrm{SNR}_{\mathrm{linear},1} = 272.25, \qquad \mathrm{SNR}_{dB,1} = 20\log_{10}(272.25) = 48.70 \text{ dB} \tag{3}$$

$$\mathrm{SNR}_{\mathrm{linear},8} = 773.22, \qquad \mathrm{SNR}_{dB,8} = 20\log_{10}(773.22) = 57.77 \text{ dB} \tag{4}$$

$$\mathrm{SNR}_{\mathrm{linear},128} = 3908.32, \qquad \mathrm{SNR}_{dB,128} = 20\log_{10}(3908.32) = 71.84 \text{ dB} \tag{5}$$

$$\mathrm{SNR}_{\mathrm{linear},4096} = 15169.31, \qquad \mathrm{SNR}_{dB,4096} = 20\log_{10}(15169.31) = 83.62 \text{ dB} \tag{6}$$

Measured SNR improvements

$$\Delta\mathrm{SNR}_{1\to8} = 57.77 - 48.70 = 9.07 \text{ dB} \tag{7}$$

$$\Delta\mathrm{SNR}_{8\to128} = 71.84 - 57.77 = 14.07 \text{ dB} \tag{8}$$

$$\Delta\mathrm{SNR}_{128\to4096} = 83.62 - 71.84 = 11.78 \text{ dB} \tag{9}$$

$$\Delta\mathrm{SNR}_{1\to4096} = 83.62 - 48.70 = 34.92 \text{ dB} \tag{10}$$

Averaging Equation:

$$\Delta\mathrm{SNR}_{dB} = 10\log_{10}(N)[1] \tag{11}$$

So from no averaging to $N = 4096$:

$$\Delta\mathrm{SNR}_{\mathrm{theory}} = 10\log_{10}(4096) = 36.12 \text{ dB} \tag{12}$$

Compared to measured:

$$\Delta\mathrm{SNR}_{\mathrm{measured}} = 34.92 \text{ dB} \tag{13}$$

Total improvement from no averaging to 4096 averages:

$$\Delta\mathrm{SNR} = 34.92\,\mathrm{dB}$$

The results for SNR show an increasing trend as the number of averages is increased. As expected, the measured SNR increases from 48.70 dB for no averages to 83.62 dB for 4096 averages, an increase of 34.92 dB. This matches the expected behavior for the number of averages

$$\Delta\text{SNR} = 10\log_{10}(N)$$

which yields 36.12 dB for $N = 4096$. The minor difference between the calculated and ideal values is attributed to random noise components, equipment drift, and quantization. Averaging is effective in improving the SNR, as indicated in Figure [2].

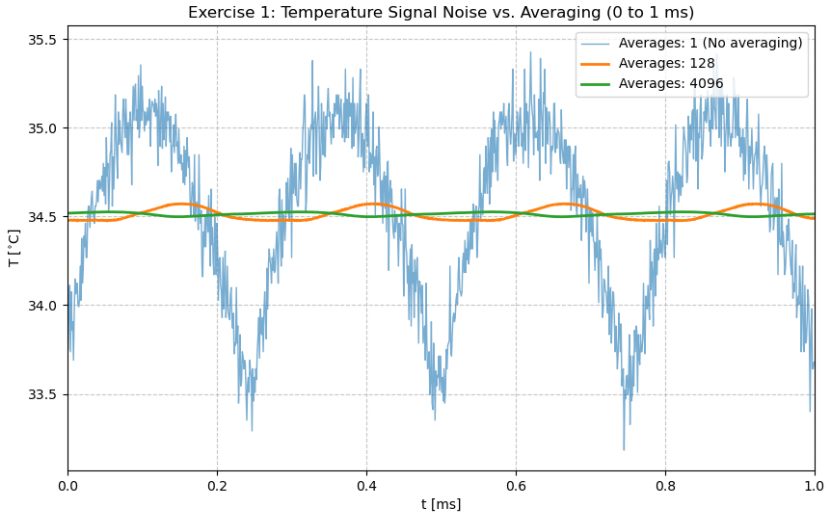


Figure 2: The Temperature Signal noise Vs Averaging (0 to 1 ms)

2.2 FFT Noise Analysis (Low -> Mid -> High Peaks)

- Low-frequency peak: 4.0 kHz
- Mid-frequency peak: 95.4 kHz
- High-frequency peak: 179.9 kHz

The FFT analysis can be used to identify the major frequency components in the noise. Low frequency can be caused by a drift or environmental fluctuations Figure [3], whereas mid and high frequency can be caused by electronic interference Figure [4]. It becomes possible to differentiate between random noise and periodic interference components in the noise after performing an FFT analysis.

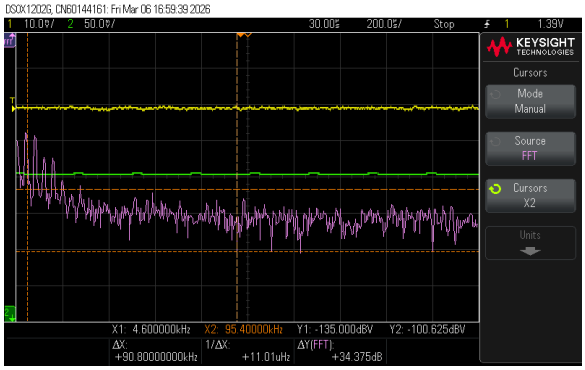


Figure 3: FFT Spectrum $X_1(F_{low}) = 4kHz$,$X_2 = F_{mid} = 95kHz$

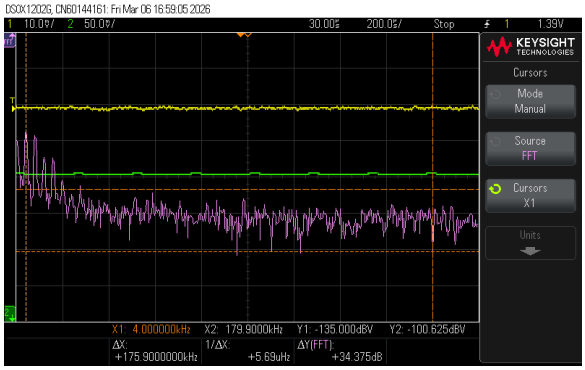


Figure 4: FFT Spectrum $X_1(F_{High}) = 179.9kHz$

2.3 Bandwidth Limit Effect (BW OFF vs BW ON)

Turning the oscilloscope's bandwidth limit on should minimize the effect of the noise components at these higher frequencies. This should result in a smoother waveform displayed on the oscilloscope along with a reduction in the amplitude of the noise. This proves that the noise level is not in the frequency range that it needs to be in to measure the temperature level. This observation cannot be noticeable on the oscilloscope; it could be because it is minimal Figure [6]. A slight increase in amplitude was noted in the oscilloscope's reading at the second low frequency peak after turning on the oscilloscope's bandwidth limit. The peak amplitude in the range of 0 to 200 kHz changed slightly, as shown in Figure [5]. This indicates that there is no effect of bandwidth limiting in that range.

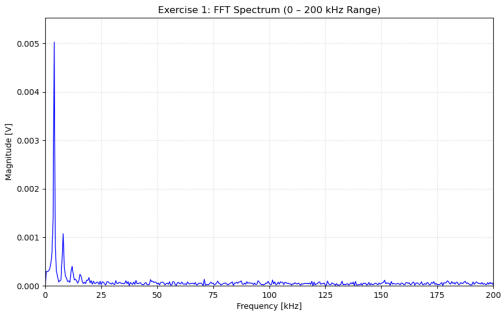


Figure 5: BW limits CSV data

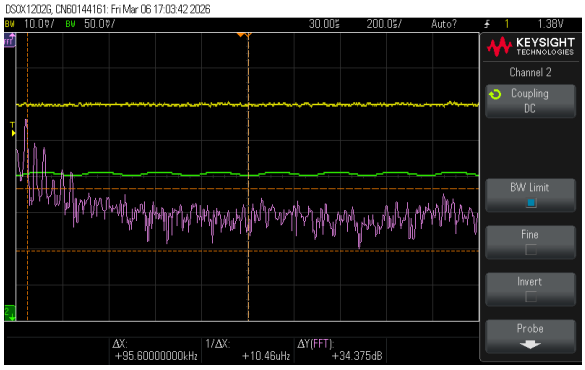


Figure 6: BW limits OSC. data

2.4 Capacitor Filtering Effect

The capacitor passive filtering test was presented as a controlled comparison of three cases:

1. No capacitor (baseline),Figure [9]

2. 10 µF capacitor,Figure [7]

3. 47 µF capacitor,Figure [8]

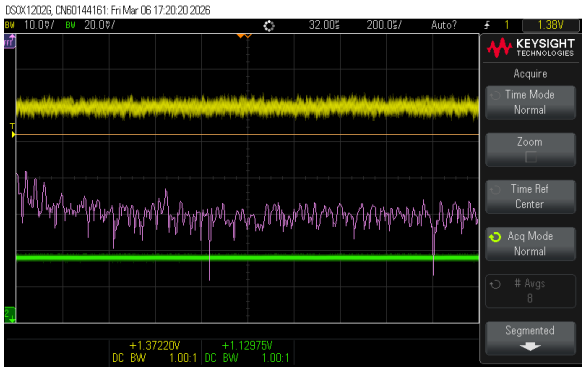


Figure 7: 10 µF capacitor

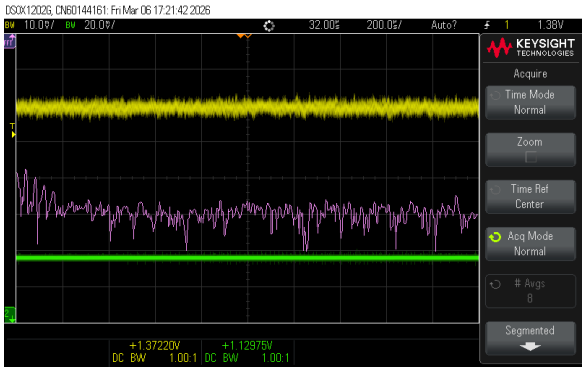


Figure 8: 47 µF capacitor

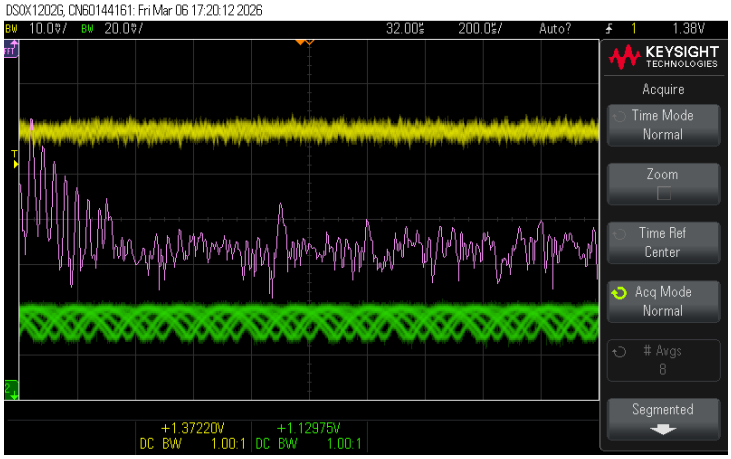


Figure 9: no-capacitor

The addition of a capacitor in the amplutide output increases the signal by decreasing the noise compared to the no capacitor case. The 47 µF capacitor provided a stronger smoothing effect than the 10 µF capacitor.

2.5 Sampling Rate and Nyquist

The sampling rate will define how accurately the oscilloscope will take the signal in time. In order for the sampling not to produce an aliased waveform, the sampling frequency has to be at least double the highest frequency present in the waveform[+ref]; according to the Nyquist definition. If not, the frequency components may not show correctly.

The sampling rate section must use consistent definitions:

$$f_s = \frac{1}{T_s}[2] \tag{14}$$

$$f_s \geq 2f_{\max}[2,3] \tag{15}$$

where f_s is sampling rate (Hz) and T_s is sampling period (s).

Case 1: if $f_{\max} = 4\,\text{kHz}$ Nyquist condition:

$$f_s \geq 2 \times 4\,\text{kHz} = 8\,\text{kHz} \tag{16}$$

Equivalent period limit:

$$T_s \leq \frac{1}{8000} = 125\,\mu\text{s} \tag{17}$$

So $T_s = 100\,\mu\text{s}$ is valid only for $4\,\text{kHz}$.

Case 2: using measured FFT peaks Based on the measured spectrum, there were peaks at 95.4 kHz and 179.9 kHz. Thus, the maximum observed frequency was:

$$f_{\max} = 179.9\,\text{kHz}.$$

Nyquist requirement:

$$f_s \geq 2 \times 179.9\,\text{kHz} = 359.8\,\text{kS/s} \tag{18}$$

Equivalent maximum sampling period:

$$T_{s,\max} = \frac{1}{359.8 \times 10^3} = 2.78\,\mu\text{s} \tag{19}$$

Therefore, $T_s = 100\,\mu\text{s}$ ($f_s = 10\,\text{kHz}$) would alias these high frequency components.

With a maximum noise frequency of 179.9 kHz as determined, the minimum sampling rate is 359.8 kHz ($T_{s,\max} = 2.78\,\mu\text{s}$). It is therefore necessary for the sampling rate to be higher than 10 kHz unless it is assumed that the high frequency components of the signal have been limited by filtering.

With ($f_{\max} = 4\,\text{kHz}$), it is assumed that the oscilloscope configuration is valid. With $f_s \geq 8\,\text{kHz}$as determined by the Nyquist criterion, and ($T_s \leq 125\,\mu\text{s}$), it is valid since $T_s = 100\,\mu\text{s}$.

2.6 Quantization Error Across Five Sensitivity Settings

The quantization error is a result of the conversion of analogue voltage signals to digital voltage signals by the oscilloscope, The voltage resolution becomes wider for low sensitivity data and this may cause an increase in the quantization step size. The quantization resolution becomes high sensitivity data and this allows for the measurement of smaller voltage variations. This is especially seen in low amplitude sensor signals look Table[1].

Sensitivity data (.CSV)	(µV RMS)	Peak-to-peak (µV)
ex4_2	1.447	11.730
ex4_1	2.723	19.050
ex23_1_2	199.828	1401.280
ex4_0	210.883	1694.820
ex23_0_2	2071.928	12503.420

Table 1: Quantization/noise comparison for five selected sensitivity settings.

Conclusion Quantization noise level increases as one moves from one sensitivity setting to another.The least quantization noise level was recorded for `ex4_2` (1.447 µV RMS), and the highest quantization noise level was recorded for `ex23_0_2` (2071.928 µV RMS), showing a clear increase in effective quantization noise with lower sensitivity data, Figure [10].

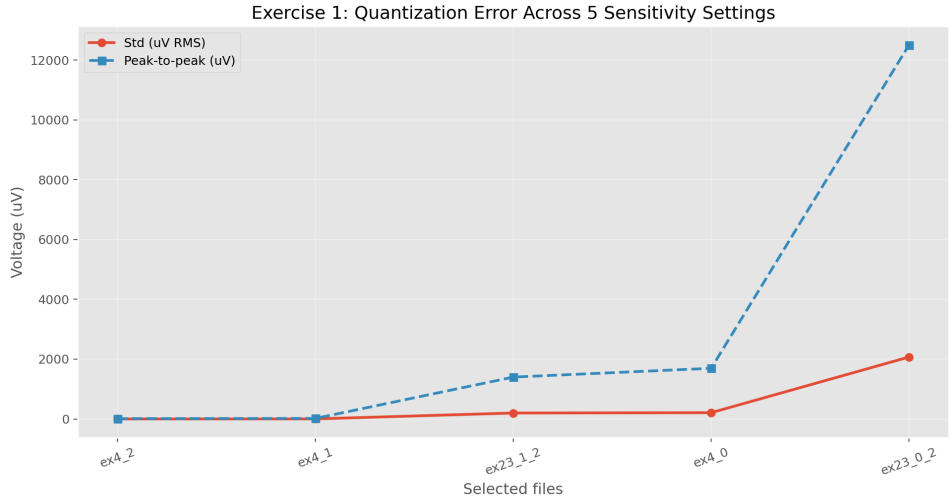


Figure 10: Uantization 5 settings

3 Exercise 2: RTD and Thermopile Temperature Measurements

This section analyzes temperature measurements from RTD and thermopile sensors during heating and cooling cycles. The two sensors were measured in parallel under the same thermal process and oscilloscope acquisition conditions, so their trend and agreement can be compared directly. The objectives are to evaluate heating/cooling behavior, estimate the thermopile sensitivity coefficient, assess RTD linearity, calculate room temperature random uncertainty, and examine histogram based distribution differences.

3.1 Heating and cooling results

Both sensors followed the same heating/cooling trend (high correlation, $r \approx 0.995$ in both cycles), but agreement was cycle dependent, Figure[11].

- rep2: RTD 36.79 °C to 69.54 °C (span 32.75 °C); thermopile 29.49 °C to 74.01 °C (span 44.52 °C); thermopile−RTD −7.30 °C to 4.47 °C; MAE = 4.16 °C.
- rep3: RTD 36.13 °C to 69.20 °C (span 33.07 °C); thermopile 28.84 °C to 49.91 °C (span 21.08 °C); thermopile−RTD −19.29 °C to − 7.29 °C; MAE = 10.61 °C.

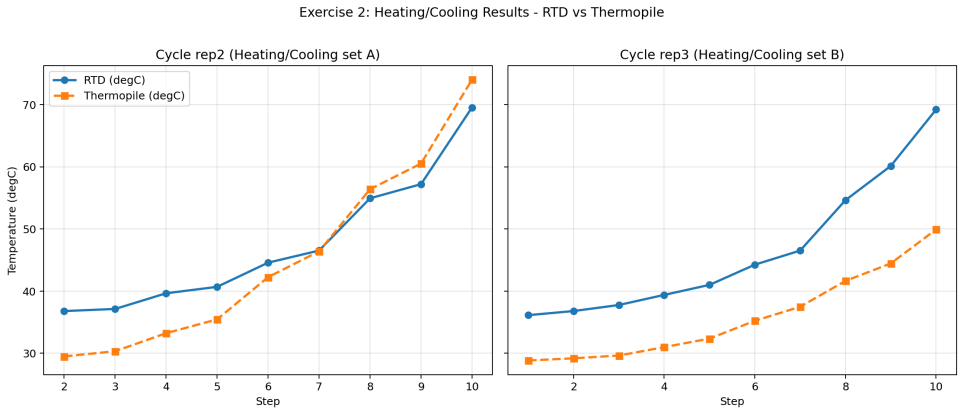


Figure 11: Heating / Cooling RTD vs Thermopile

3.2 Thermopile sensitivity coefficient and RTD linearity error

Thermopile sensitivity was estimated by linear fit:

$$V_{\text{thermopile}} = S\,T_{\text{RTD}} + V_0[4,5] \tag{20}$$

Results:

- rep2: $S = 6.9603\,\text{mV}/°\text{C}$, $V_0 = 1.146\,460\,\text{V}$, $R^2 = 0.99114$
- rep3: $S = 3.2283\,\text{mV}/°\text{C}$, $V_0 = 1.279\,478\,\text{V}$, $R^2 = 0.99037$
- combined: $S = 5.0387\,\text{mV}/°\text{C}$, $V_0 = 1.215\,272\,\text{V}$, $R^2 = 0.76567$

So overall sensitivity is about $5.04\,\text{mV}/°\text{C}$, with strong run-to-run variation.

RTD linearity was evaluated with BFSL:

$$T_{\text{RTD}} = a\,T_{\text{ref}} + b[5] \tag{21}$$

$$e_L = \max|T_{\text{RTD}} - T_{\text{fit}}|\,[4] \tag{22}$$

Max BFSL linearity error:

- rep2: 1.680 °C (5.13 %FS)
- rep3: 2.373 °C (7.18 %FS)

(Combined fit is poor due to cycle mismatch, Figure12.)

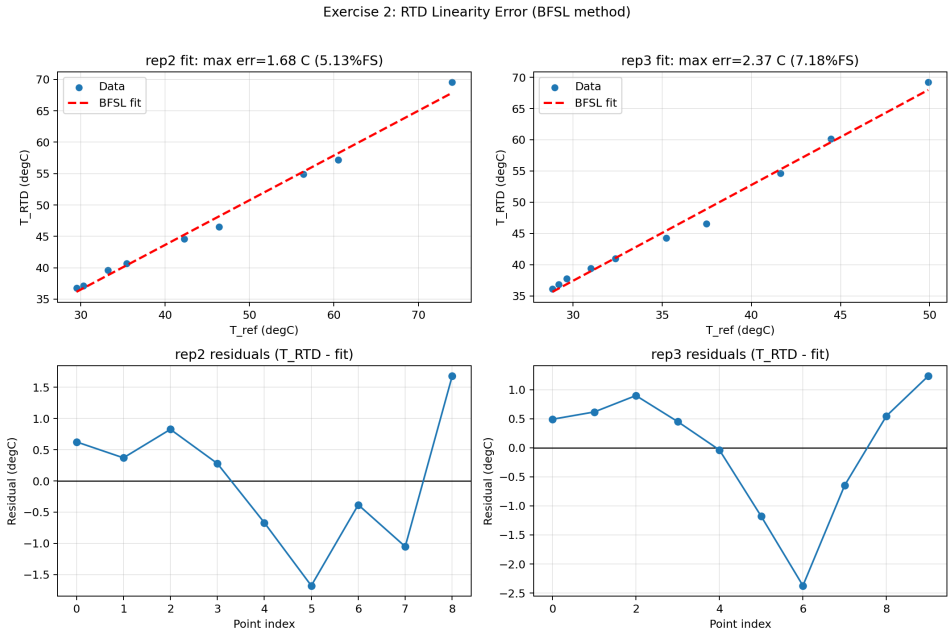


Figure 12: RTD linearity error

3.3 Random uncertainty at room temperature

Room-temperature data: `ex23_1_2` and `ex23_1_3` ($n = 4000$ combined). Type-A method:

$$u_A = \frac{s}{\sqrt{n}}[6] \tag{23}$$

$$U_{95} = 1.96\, u_A[7] \tag{24}$$

- Thermopile: mean 28.643 °C, $s = 0.198$ °C, $u_A = 0.0031$ °C, $U_{95} = \pm 0.0061$ °C
- RTD: mean 36.288 °C, $s = 0.184$ °C, $u_A = 0.0029$ °C, $U_{95} = \pm 0.0057$ °C

3.4 Histograms and normality/distribution differences

Raw combined histograms from both runs are mixture distributions and can appear bimodal due to run to run mean shift. Therefore, normality/noise comparison should use per-run histograms and de-meaned combined residuals.

- Per run: both sensors are approximately unimodal, Figure [13].

- De-meaned pooled spread: thermopile $\sigma = 0.0406\,^{\circ}\mathrm{C}$, RTD $\sigma = 0.0925\,^{\circ}\mathrm{C}$, Figure [14].
- Interpretation: neither is perfectly Gaussian, but both are close to unimodal noise like distributions after removing run offsets; RTD shows larger within run spread in this dataset.

Conclusion RTD and thermopile track the same thermal process, but absolute agreement and inferred thermopile sensitivity depend on cycle/setup. RTD is approximately linear within each cycle ($\sim 5\,\%$FS to $7\,\%$FS), room temperature Type A uncertainty is small for both sensors, and histogram interpretation must account for run offset effects.

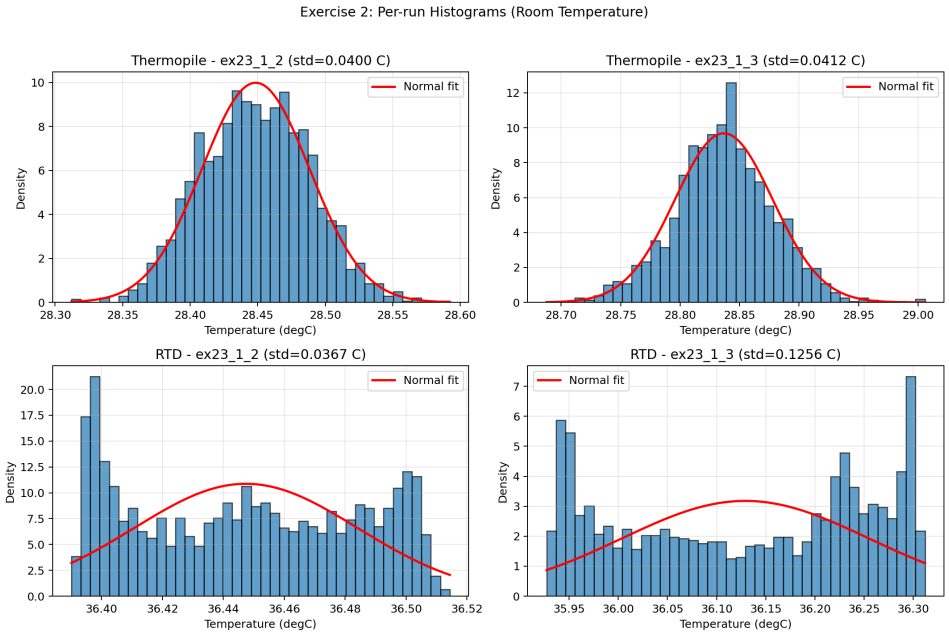


Figure 13: Histograms per Run Roomtemp

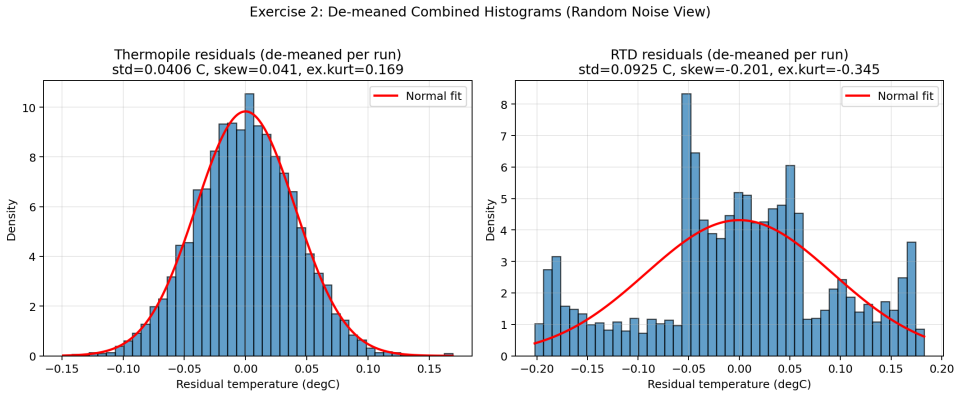


Figure 14: Histograms de-meaned Combined

4 Exercise 3: Floating Thermocouple and RTD

This section analyzes temperature measurements from a floating thermocouple and an RTD over multiple temperature points. These measurements were performed in parallel with Exercise 2 (same thermal process and acquisition setup), enabling direct comparison of sensor behavior under similar conditions. The goals are to quantify thermocouple error relative to RTD, estimate thermocouple sensitivity, compare fitted parameters with the reference model, test agreement within uncertainty bounds, and discuss uncertainty sources for both configurations as in Figure 15

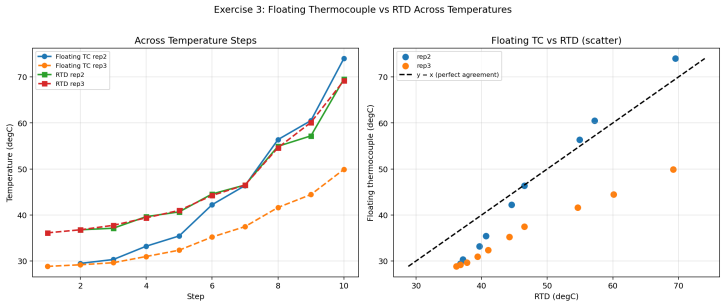


Figure 15: Floating Thermocouple vs RTD

4.1 Key formulas

$$e = T_{\mathrm{TC}} - T_{\mathrm{RTD}}[8] \tag{25}$$

$$V_{\mathrm{TC}} = S\,T_{\mathrm{RTD}} + V_0[4, 9] \tag{26}$$

$$|T_{\mathrm{TC}} - T_{\mathrm{RTD}}| \leq U_{95,\mathrm{combined}}[8] \tag{27}$$

4.2 Error and sensitivity results

repet 2 (better run)

- TC range: 29.49 to 74.01 °C
- RTD range: 36.79 to 69.54 °C
- Error e: -7.30 to 4.47 °C
- Correlation: $r = 0.9956$
- MAE: 4.16 °C, RMSE: 4.79 °C, Bias: -2.11 °C
- Sensitivity fit: $S = 6.960\,\mathrm{mV/°C}$, $V_0 = 1.146460\,\mathrm{V}$, $R^2 = 0.9911$
- Sensitivity deviation vs 5.00 mV/°C: +39.21%

repet3 (worse run)

- TC range: 28.84 to 49.91 °C
- RTD range: 36.13 to 69.20 °C
- Error e: -19.29 to -7.29 °C
- Correlation: $r = 0.9952$
- MAE: 10.61 °C, RMSE: 11.28 °C, Bias: -10.61 °C
- Sensitivity fit: $S = 3.228\,\text{mV/°C}$, $V_0 = 1.279478\,\text{V}$, $R^2 = 0.9904$
- Sensitivity deviation vs 5.00 mV/°C: -35.43%

Combined

- Correlation: $r = 0.8750$
- Error e: -19.29 to 4.47 °C
- MAE: 7.56 °C, RMSE: 8.82 °C, Bias: -6.58 °C
- Fit: $S = 5.039\,\text{mV/°C}$, $V_0 = 1.215272\,\text{V}$, $R^2 = 0.7657$
- Look Figure 16



Figure 16: Floating Thermocouple vs RTD

4.3 Comparison with reference model

Reference:

$$V_{\mathrm{TC,ref}} = 0.005\,T + 1.25[9]$$

($S_{\mathrm{ref}} = 5.00\,\mathrm{mV/°C}$, $V_{0,\mathrm{ref}} = 1.250\,\mathrm{V}$).

Simple meaning

- rep2 is too high in sensitivity.
- rep3 is too low in sensitivity.
- combined sensitivity is close to nominal, but per-run temperature errors are still large.
- Look Figure 17

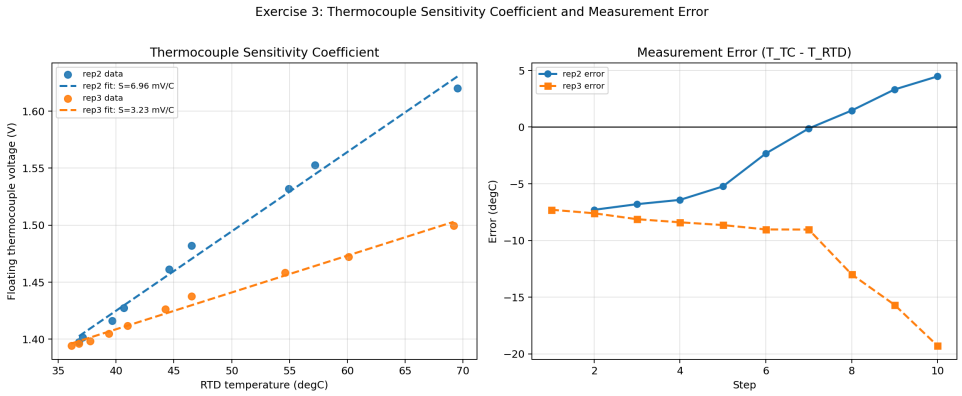


Figure 17: Floating Thermocouple vs RTD

4.4 Do the TC and RTD agreed?

Overall the two sensors do **not** agree within estimated uncertainty for this dataset.

4.5 Main uncertainty sources

- Sensors not exactly at same location
- Temperature gradients and transient lag/hysteresis
- Quantization/noise and averaging effects
- Repeatability (airflow, handling, pressure/contact)

4.6 Final conclusion

RTD and floating thermocouple move together in trend, but they do not match well in absolute temperature. Results change a lot between runs, which points to setup/calibration systematic effects (gain, offset, reference,placement), not only random noise.

5 Exercise 4: Limited Cases (Open Circuit / Disconnected Sensors)

When a sensor is disconnected, the measurement is not a real temperature anymore. This is a limiting fault case.

Interpretation If RTD or thermocouple is open-circuit, the input becomes floating (high impedance). Then the electronics (bias currents, leakage, and input circuitry) push the signal to a default voltage level. After conversion, this gives a temperature value that may look stable, but it is not physical.

Why this matters So, disconnected sensor readings must be treated as invalid/fault data, not true temperature measurements.

Observed open–circuit results From open-circuit datasets (`ex4_1`, `ex4_2`):

- CH1 $\approx 1.444\,37\,\mathrm{V} \Rightarrow \approx 38.87\,^{\circ}\mathrm{C}$ using thermocouple conversion.
- CH2 $\approx 1.852\,46\,\mathrm{V} \Rightarrow \approx 122.30\,^{\circ}\mathrm{C}$ using RTD conversion.

Variation is very small, indicating a bias–defined level rather than real thermal response. In `ex4_0`, noise increases, but readings are still centered around these bias-driven levels.

Conclusion Open-circuit outputs must be treated as **fault-state / invalid measurements**, not valid temperature data.

6 Discussion:

The results demonstrate that averaging can greatly enhance the SNR, as expected from statistical reduction of random noise. FFT analysis can be employed to differentiate broadband noise from discrete spectral peaks. In terms of bandwidth limiting there were minimal changes in the signal in the 0–200 kHz range with a small rise in low frequencies. Therefore, no significant in band effect was observed from the results; quantization/noise

spread observed when the sensitivity of the oscilloscope was not well matched to signal amplitude. In terms of sensor comparison although the RTD and thermopile/floating thermocouple responded to the same thermal trend, there were some discrepancies in the results. RTD linearity was good within each individual cycle, although sensitivity and offset of the thermopile/floating thermocouple varied significantly from run to run. Uncertainty was metric dependent; RTD had slightly lower Type A U_{95}, although de–meaned pooled spreads were greater for RTD in the results at room temperature.

References

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